

HAMZA MOORAJ

Machine Learning Engineer — Multimodal & Agentic Systems | ML Systems

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AI engineer focused on building and deploying multimodal and agentic systems for real-world applications. Experience designing end-to-end ML pipelines, integrating retrieval and model components, and optimising system performance under practical constraints.

PROJECTS

A Systematic Exploration of Architectural Trade-offs for Crop Disease Classification - Lead Author

July 2025 – April 2026 — Under Review at **TLMR** (2026) — [Dataset released on HuggingFace](#) — [arXiv Preprint](#)

- Built a multi-source dataset pipeline (12 datasets → unified 65-class schema with lab/field splits).
- Implemented training workflows across CNN and VLM architectures with configurable experimentation.
- Developed evaluation pipelines for cross-domain testing and metric tracking (F1, precision, recall).
- Integrated experiment tracking and reproducibility tooling using Weights & Biases.

Medical Question-Answering (QA) Conversational Agent (Course: F21CA) - Project Lead

January 2025 – April 2025

- Built a RAG-based medical QA system over 411 NHS documents using FAISS retrieval.
- Designed a tri-modal pipeline integrating text, speech, and vision components.
- Implemented retrieval and generation workflows for improved response accuracy.
- Evaluated system performance and improved output quality via retrieval grounding (+5.6%).

Mechanistic Analysis of Induction in a Tiny Transformer - Personal Project

February 2026

- Implemented and trained a 2-layer transformer model for controlled sequence prediction tasks.
- Built tooling to visualise and analyse attention patterns across model layers and heads.
- Developed intervention workflows (ablation, perturbation) to test model behaviour under modifications.
- Designed experimental setups to study robustness of learned patterns.

EXPERIENCE

AI “Learning Buddy” Project Intern — Heriot-Watt University

July 2025 – September 2025

- Built an end-to-end retrieval-based learning assistant using LlamaIndex and Milvus.
- Implemented ingestion, chunking, and metadata-aware indexing pipelines for course materials.
- Designed retrieval workflows to improve relevance across heterogeneous educational content.
- Deployed the system in a containerised environment and exposed a live demo for testing.

Data Science & Machine Learning Intern — Love For Data

August 2023

- Built a data pipeline and popularity index across 1,970 Play Store apps (US and Pakistan datasets).
- Trained and evaluated a Random Forest model to identify key performance drivers.

Software Engineering Intern — Folio3

June 2023 – July 2023

- Contributed to a production Visitor Management System within a sprint-based development team.

EDUCATION

Heriot-Watt University — BSc Computer Science (Artificial Intelligence)

First Class Honours, 3x Deputy Principal's Award

BlueDot Impact — Technical AI Safety Course (2026) — [Credentials](#)

SKILLS

Languages: Python, Java, C#, TypeScript, C

Machine Learning: Transformers, Vision-Language Models, PEFT (LoRA), RAG

Systems & Infrastructure: PyTorch, Milvus, Docker, Linux, CUDA

Tools: Git, Weights & Biases, RunPod, LangGraph